



Ask a Question



Understanding DNS – Part 1

In this lesson, you will learn what the domain name system (DNS) is and how it works.

We'll cover the following



- Domain name system
- How does a domain name system work?



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the web using that particular *IP address*.

IP stands for *Internet Protocol*. It's a protocol that facilitates the delivery of data packets from one machine to another using their IP addresses.

2001:db8:0:1234:0:567:8:1 - This is an example of a machine's *IP address*. The server that hosts our website will have a similar *IP address*. To fetch content from that server, a user has to type in the unique IP address of the server in their browser's URL tab and hit enter to interact with the website's content.

Naturally, it is not viable to type in the website's IP address from memory every time we visit a particular website. Even if we try to, how many IP addresses do you think you can remember?



Typing in domain names, for instance, [amazon.com](https://www.amazon.com), is a lot easier than working directly with IP addresses. I think we can all agree on this.



Domain name system

Domain name system, commonly known as *DNS*, is a system that averts the need to remember long *IP addresses* to visit a website by mapping easy-to-remember domain names to IP addresses.



amazon.com is a domain name that is mapped to its unique *IP address* by the *DNS* so that we are not expected to type in the *IP address* of *amazon.com* into our browsers every time we visit that website.

Now let's explore how DNS works?

How does a domain name system work?

When a user types in the *URL* of the website in their browser and hits enter, this event is known as *DNS querying*.

Four key components, or a group of servers, make up the DNS infrastructure. These are:

- *DNS Recursive nameserver aka DNS Resolver*
- *Root nameserver*
- *Top-Level Domain nameserver*
- *Authoritative nameserver*



